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An integrated system for publishing environmental observations data

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Abstract

Over the next decade, it is likely that science and engineering research will produce more scientific data than has been created over the whole of human history. The successful use of these data to achieve new scientific breakthroughs will depend on the ability to access, integrate, and analyze these large datasets. Robust data organization and publication methods are needed within the research community to enable data discovery and scientific analysis by researchers other than those that collected the data. We present a new method for publishing research datasets consisting of point observations that employs a standard observations data model populated using controlled vocabularies for environmental and water resources data along with web services for transmitting data to consumers. We describe how these components have reduced the syntactic and semantic heterogeneity in the data assembled within a national network of environmental observatory test beds and how this data publication system has been used to create a federated network of consistent research data out of a set of geographically decentralized and autonomous test bed databases.

Keywords

Cyberinfrastructure; Data publication; Water resources data; Web services; Data model; Semantic heterogeneity; Controlled vocabulary; Data intensive science

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Introduction

New technology and data resources are often instrumental in the emergence of new scientific discoveries. Because results from local research projects can be aggregated across sites and times, in many cases by investigators other than those who originally collected the data, the potential exists to advance science and research significantly through the publication of research data (Borgman et al., 2007, Research Information Network, 2008). There is a need, therefore, for standardized and robust methods to organize and publish environmental observations data as resources that can be discovered and used for scientific analysis.

Indeed, environmental research and education have recently become increasingly data-intensive as a result of the proliferation of digital technologies, instrumentation, and pervasive networks through which data are collected, generated, shared, and analyzed (National Science Foundation, 2007). Over the next decade, it is likely that science and engineering research will produce more scientific data than has been created over the whole of human history (Cox et al., 2006). Successfully using these data to achieve new scientific breakthroughs and increase understanding of the world around us, as well as in making sound and informed resource management decisions, will depend in large part on the ability to access, organize, integrate, and analyze these large datasets.

Comprehensive infrastructure that is being used to capitalize on dramatic advances in information technology has been termed "cyberinfrastructure" and integrates hardware for computing, data and networks, digitally enabled sensors, observatories and experimental facilities, and an interoperable suite of software and middleware services and tools (National Science Foundation, 2007). This paper describes new cyberinfrastructure that enables the publication of point observations (i.e., measurements made at a point in space such as a weather station or water quality monitoring site). This cyberinfrastructure has been developed as part of a Hydrologic Information System (HIS), which is a distributed network of data sources and functions that are integrated using web services and that provide access to data, tools, and models that enable synthesis, visualization, and evaluation of hydrologic system behavior (http://his.cuahsi.org ↗). Although the data publication system described in this paper has been developed primarily to advance the information science knowledge base and available data resources for water resources research, the general system architecture could be extended to many other types of point observations.

The HIS consists of four major components: data publication, data curation, data discovery, and data delivery. Publication is the process by which data are made available to users other than those that collected the data. Curation is the long term preservation of data to ensure that they persist indefinitely. Discovery involves tools that allow users to find published data, and delivery involves the transmittal of data to users in formats that they can use. In this paper, we focus mainly on the data publication component, although we include some discussion of the other components to place data publication in the context of the overall HIS.

Publication of research data involves persistent storage, management, and communication of data to potential users. Within and across research sites, multiple investigators and organizations are involved in both collecting and consuming data. To be effective, data publication systems must facilitate interoperation and mediation among data sources and their consumers. One challenge that arises in the design of data publication systems is heterogeneity within the formats and vocabularies that support the data (Colomb, 1997, Morocho et al., 2003, Sheth and Larson, 1990). Additionally, data consumers may not have intimate knowledge of the data collection process, requiring that the data be published with sufficient metadata to enable unambiguous interpretation (Gray et al., 2005). These metadata should include information about the location at which the observations were made, the variable that was observed or measured, the source of or organization that created the data, the procedures used to create the data, data qualifying comments, quality assurance and quality control information, time support, spacing, and extent, and other important attributes (Horsburgh et al., 2008).

In this paper, we describe a data publication system that overcomes the challenges in publishing research data through the use of a standard observations data model populated using controlled vocabularies for environmental and water resources data along with web services for transmitting data to consumers. Section 2 describes existing data publication efforts for environmental and water resources data. Section 3 describes new cyberinfrastructure efforts ongoing within several scientific domains. Section 4 describes syntactic and semantic heterogeneity and their implications for the publication, search for, and interpretation of existing environmental and water resources data. Section 5 describes how this heterogeneity can be overcome. Section 6 describes the design of a cyberinfrastructure that supports publication of environmental observations. Finally, Section 7 provides an implementation case study that describes how the components of the data publication system have been applied to create a federated network of consistent research data out of a set of geographically decentralized and autonomous databases from 11 environmental observatory test beds, effectively creating a publically available, community data resource from data that might otherwise have been confined to the private files of the individual investigators.

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Existing data publication methods

Within the United States, many organizations and individuals measure hydrologic variables such as streamflow, water quality, groundwater levels, soil moisture, and precipitation. Several national data collection and publication networks operated by government agencies have arisen over the years. These include the USGS WATer Data STOrage and REtrieval System (WATSTORE), which has been replaced by the National Water Information System (NWIS) (http://waterdata.usgs.gov/nwis ↗), the USEPA STOrage and …

New cyberinfrastructure efforts

There are currently several large-scale cyberinfrastructure activities underway that are aimed at creating and sharing multidisciplinary datasets, facilitating collaborative and interdisciplinary research, and creating infrastructure to enable scientific discoveries. These activities include: the National Ecological Observatory Network (NEON), which is planning the deployment of networked sensors and cyberinfrastructure to gather data on compelling ecological challenges (http://www.neoninc.org ↗ …

Syntactic and semantic heterogeneity in environmental observations data

Syntactic heterogeneity refers to a difference in how data and metadata are organized (e.g., rows vs. columns) and encoded (e.g., text files versus Excel spreadsheets), while semantic heterogeneity refers to the variety in language and terminology used to describe observations. Syntactic heterogeneity arises where there are methodological inconsistencies. For example, data downloaded from automated data loggers are generally encoded as delimited text files, whereas data generated as a result of …

Overcoming heterogeneity

Reconciling heterogeneity in data from different sources, which may be required both within and across research sites, is a complex problem that has a long history in information science (Bergamaschi et al., 2001, Colomb, 1997, Cox et al., 2006). This challenge is fueling much of the movement toward using standardized markup languages as self-describing, common data formats that can be used by data producers and data consumers. Examples include Earth Science Markup Language (ESML) (Ramachandran …

Design of a cyberinfrastructure for publishing environmental observations

The objective of a cyberinfrastructure for publishing environmental observations is to enable disparate users to publish data in a way that makes them available, interpretable, and interoperable. Cyberinfrastructure that supports publication of environmental observations data must answer the following four questions: 1) what are the characteristics of data that enable them to be interpreted and how will they be persistently stored; 2) what is the mechanism by which users will access the data …

A case study for publishing point observations data: creating a national research data network

Within the United States, a network of large-scale environmental observatories, which are integrated real-time observing systems that seek to improve understanding of the earth's water and biogeochemical cycles across multiple spatial and temporal scales, has been proposed under the premise that knowledge of the physical, chemical, and biological mechanisms controlling water quantity and quality is limited by lack of observations at the necessary spatial density and temporal frequency needed to …

Discussion and conclusions

A standard method for publishing environmental and water resources point observations data has been presented. It provides a framework in which data of different types and from disparate sources can be integrated, while overcoming the syntactic and semantic heterogeneity in the data from each source. This has been the case at each site within a network of environmental observatory test beds in the United States, where publishing observational data using this system has enabled a group of …

Software and data availability

The software components described in this paper, including ODM, the ODM Data Loader, ODM Tools, the ODM controlled vocabulary system, the WaterOneFlow web services, and the central web services registry can be accessed through the CUAHSI HIS website http://his.cuahsi.org ↗, where they are distributed for free under the Open Source Berkeley Software Distribution (BSD) license. The test bed data described in this paper can be accessed through the individual web services for each test bed, which are …

Acknowledgments

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